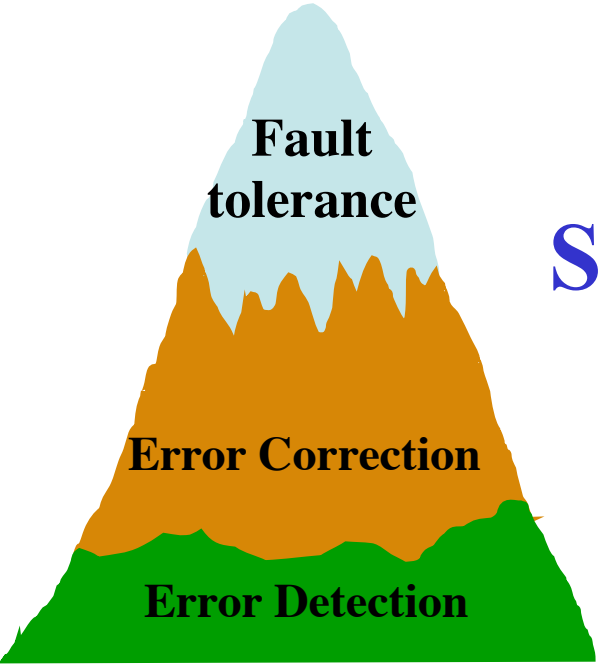
Climbing Mount Fault-Tolerance



A Trailhead:
Supercoherent Qubits

Dave Bacon

Institute for Quantum Information

California Institute of Technology

AU-US

CDEST, AAS	NSF
elong cumber colloq, mate!	elongated cumbersome colloquialisms
Coins size not correlated with value.	Coins size not correlated with value.
Vegemite	NA
ANDREW WHITE!	JONATHAN DOWLING!

Revenge slide I ✓

Rant mode ON

Computer Revolution = Physics

The Computer Science Assumption: *robust* digital storage and digital manipulation of information is possible.

In this NOISY (ANALOG) world HOW can anything be DIGITAL?

Theory: Von Neumann showed methods for fault-tolerant computation

Practice: no need for these methods because

THERE EXIST SYSTEMS WHOSE PHYSICS GUARANTEES ROBUST CLASSICAL COMPUTATION

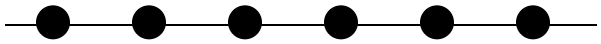
THE BILLION DOLLAR QUANTUM QUESTION

Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?

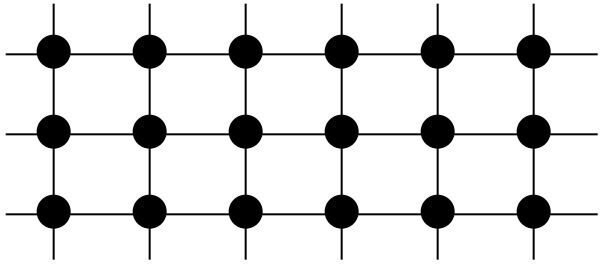
The Lesson of the Hard Drive

Classical Ising memo $\mathbf{H} = -J \sum_{\langle i,j \rangle} \boldsymbol{\sigma}_z^{(i)} \boldsymbol{\sigma}_z^{(j)}$

1D Ising model

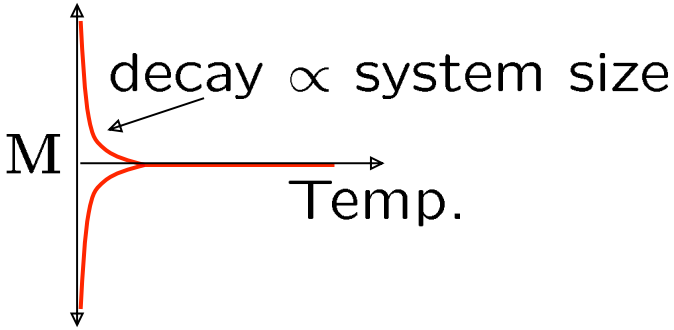


2D Ising model

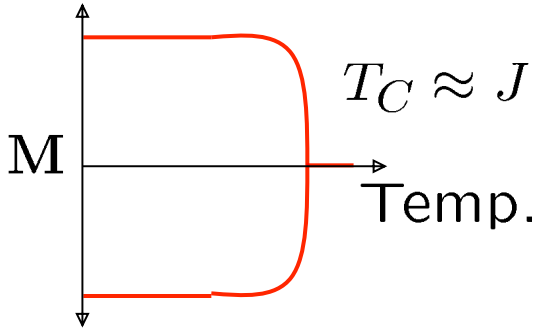


Code: all spins pointing up = 0, all spins pointing down = 1

Order parameter for this information = total magnetization $\mathbf{M} = \sum_i \boldsymbol{\sigma}_z^{(i)}$

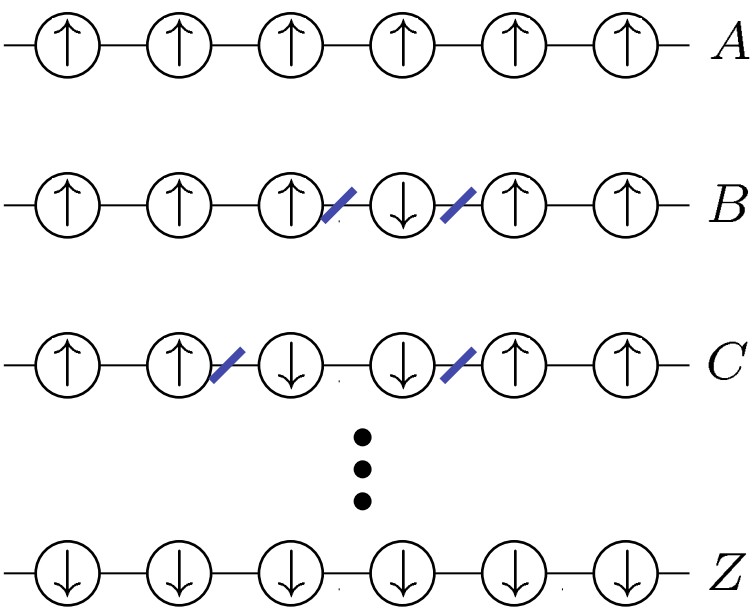


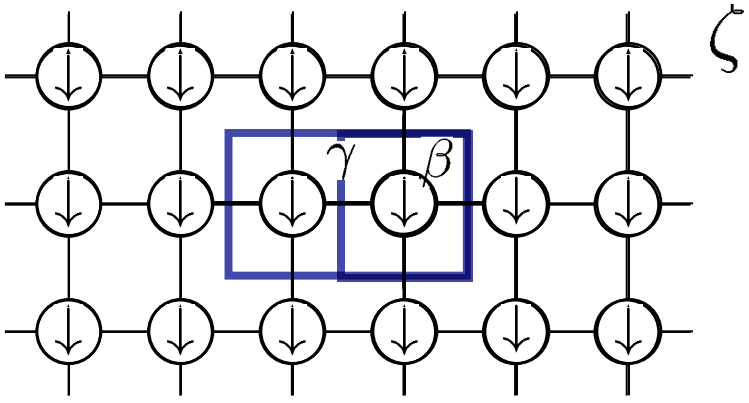
1D bad memory

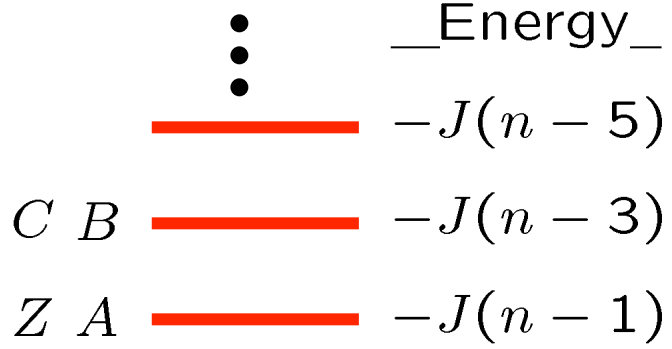


2D good memory

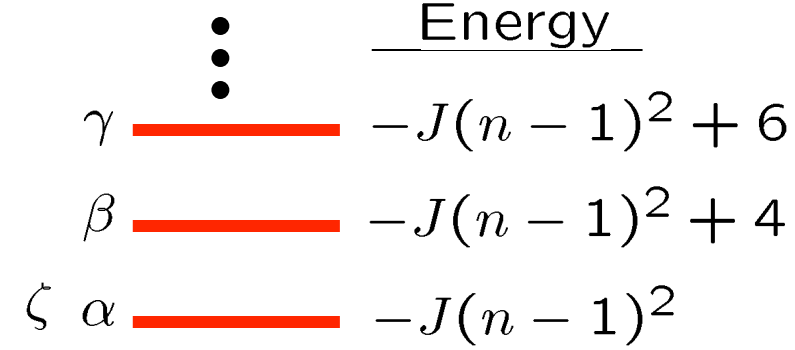
What Have We Learned?







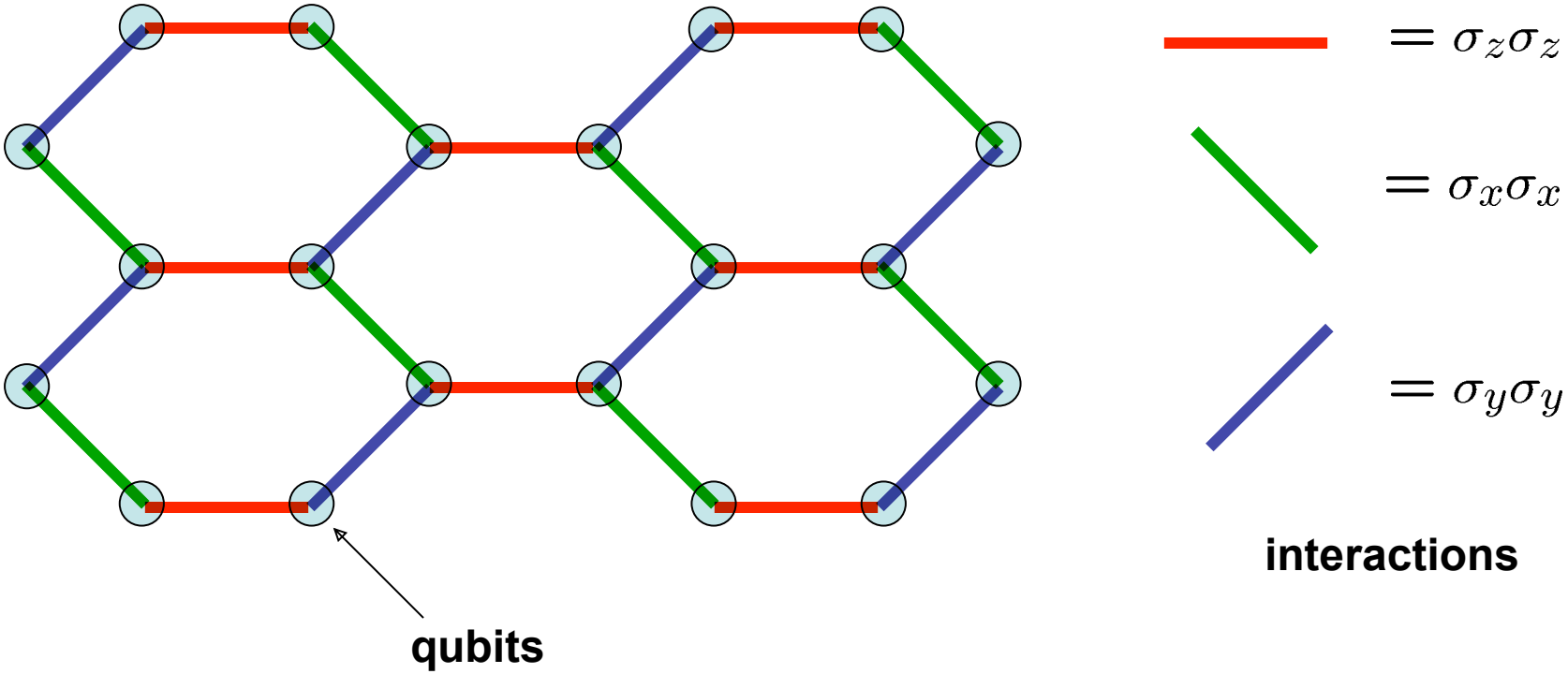
Only $O(J)$ energy used to move between codewords (error)



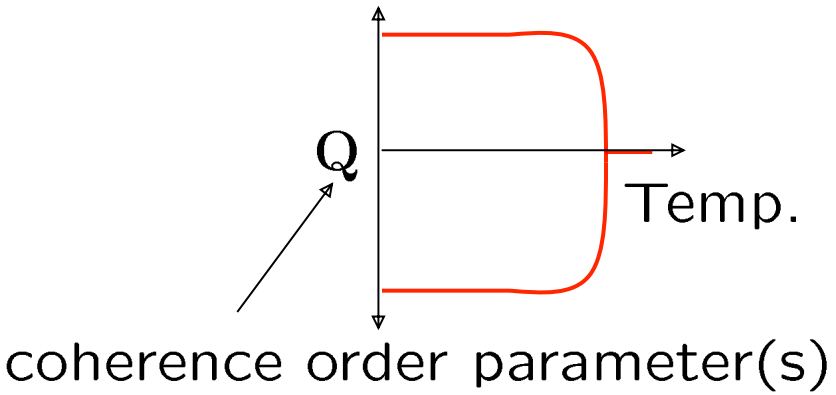
$O(nJ)$ energy used to move between codewords (error)

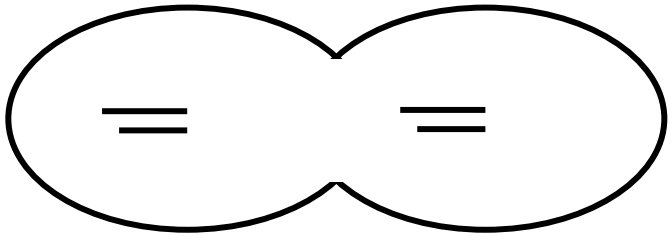
A Quantum Hard Drive?

An example of Kitaev:



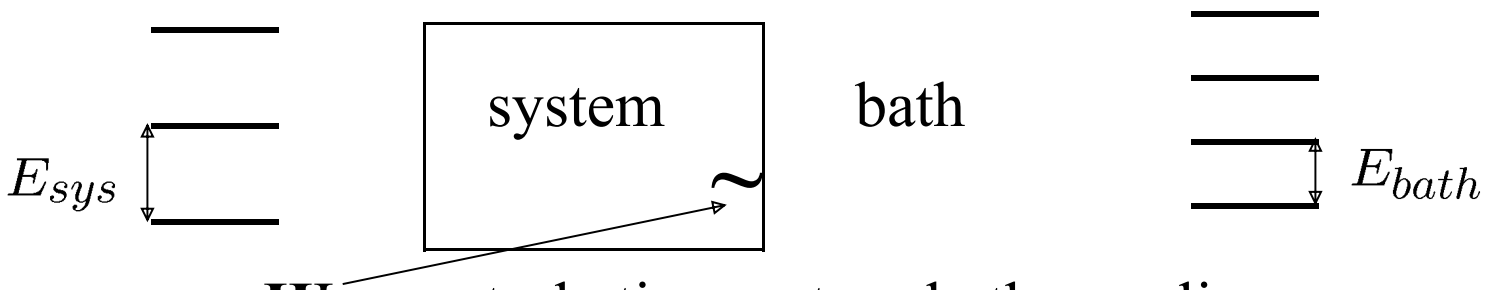
The **GOAL** of the many fancy models constructed by Kitaev and Freedman (topological quantum computation [mmmm donuts]) is to find a naturally fault-tolerant quantum computer





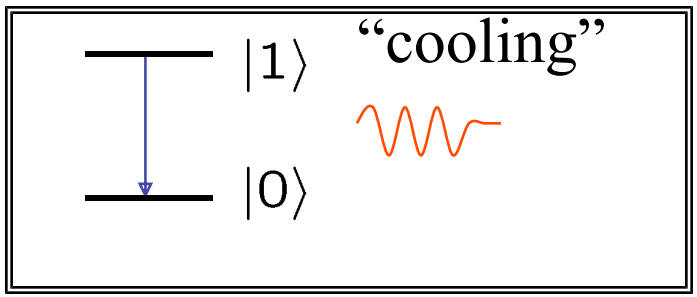
This talk: small scale example

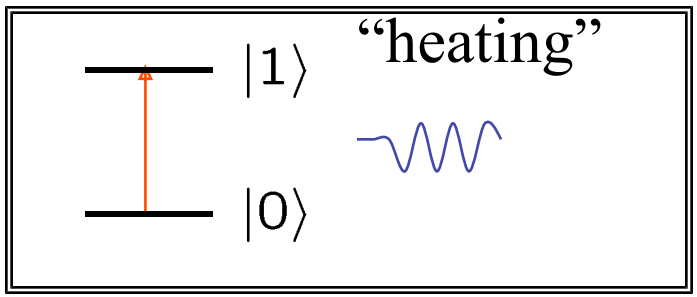
Making Decoherence Work Hard

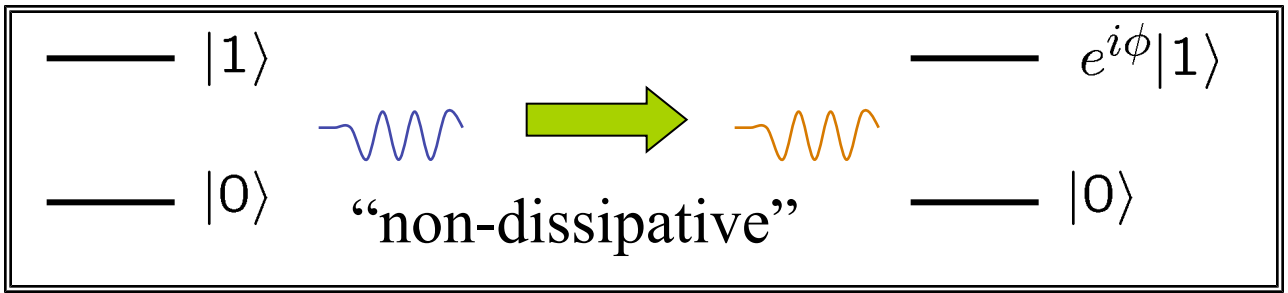


HI = perturbative system-bath coupling

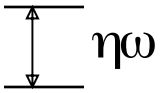
HI energy small compared to Esys and Ebath implies decoherence dominated by pathways that *conserve unperturbed energies* (essentially rotating wave approximation)







two-level atom radiatively coupled to a thermal reservoir:



$$\frac{\partial \boldsymbol{\rho}}{\partial t} = -\bar{n}_{th}\frac{\gamma}{2}\left[\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+ - 2\boldsymbol{\sigma}_+\boldsymbol{\rho}\boldsymbol{\sigma}_-\right]$$

$$-(\bar{n}_{th}+1)\frac{\gamma}{2}\left[\boldsymbol{\sigma}_+\boldsymbol{\sigma}_-\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_+\boldsymbol{\sigma}_- - 2\boldsymbol{\sigma}_-\boldsymbol{\rho}\boldsymbol{\sigma}_+\right]$$

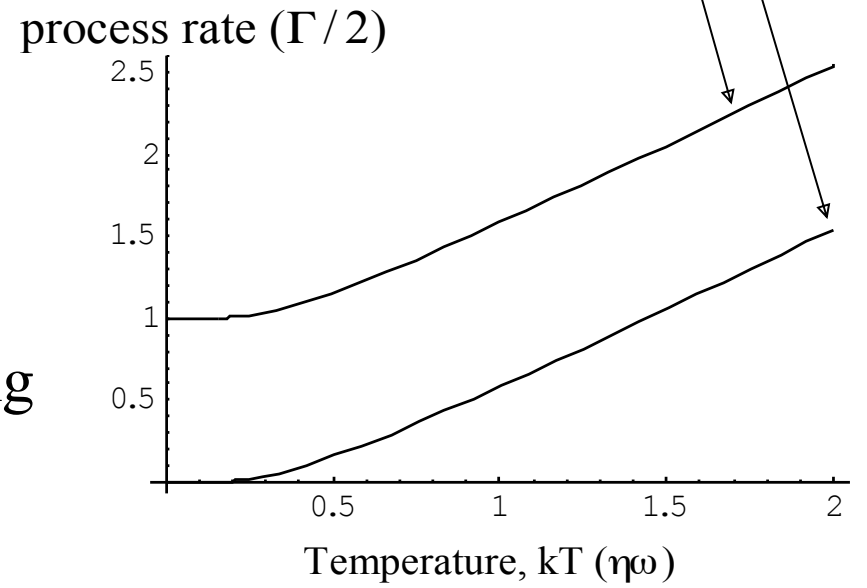
$$-\phi\left[\boldsymbol{\rho} - \boldsymbol{\sigma}_z\boldsymbol{\rho}\boldsymbol{\sigma}_z\right]$$

"heating"

"cooling"

"non-dissipative"

$$\bar{n}_{th} = \frac{1}{\exp\left(\frac{\eta\omega}{kT}\right) - 1}$$



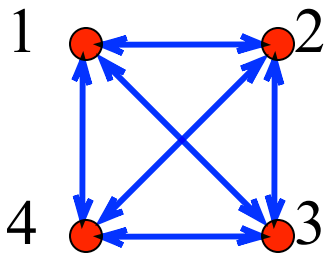
At low bath temperatures, heating disappears……..

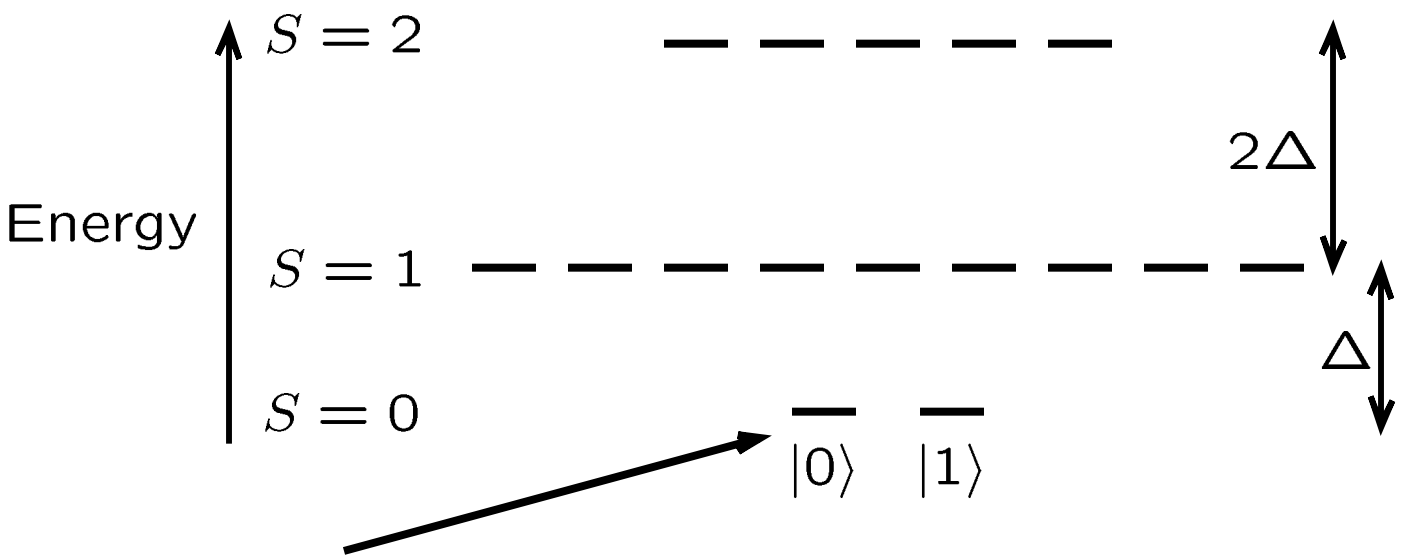
Can we make quantum errors "heating"?

A Small Example (You can build?)

Four spin systems with equal strength exchange interactions between all spins:

$$\mathbf{H} = \frac{\Delta}{2}\sum_{i,j=1}^{4} \hat{\boldsymbol{s}}_i \cdot \hat{\boldsymbol{s}}_j = \frac{\Delta}{2}\hat{\mathbf{S}}^2 + c\mathbf{I}$$





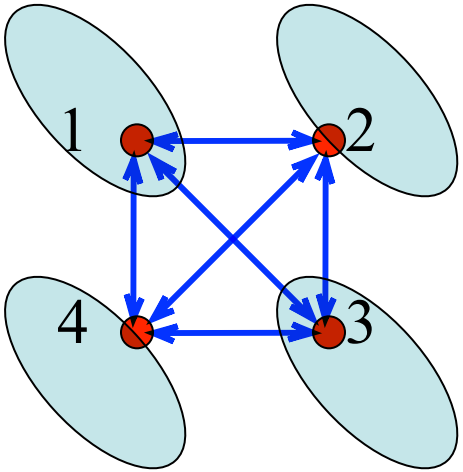
Encode a qubit of quantum information in the degenerate ground state

Stabilized by collective operators: $\mathbf{GGGG}|\psi\rangle = |\psi\rangle$

Degenerate ground states are single qubit error detecting code

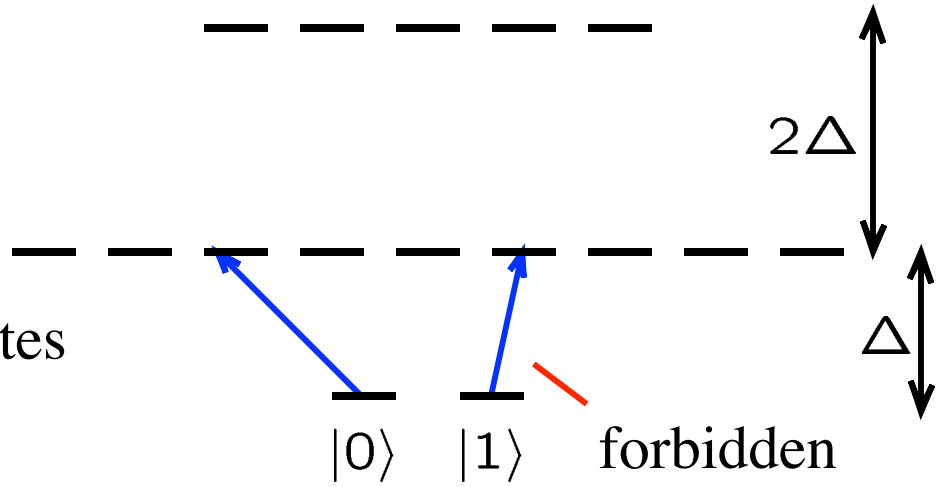
Ground state is a collective-decoherence decoherence-free subspace

Supercoherence



Qubits coupling to baths:

$$\mathbf{H}_I = \sum_{i=1}^{4} \hat{\boldsymbol{s}}_i \cdot \hat{\mathbf{B}}_i$$



All single qubit processes on this system take the ground states to states of higher energy.

As temperature of the bath is decreased $T \ll \Delta$ first-order decoherence processes will be eliminated…supercoherence

Limit to supercoherence is second order processes which act on two qubits

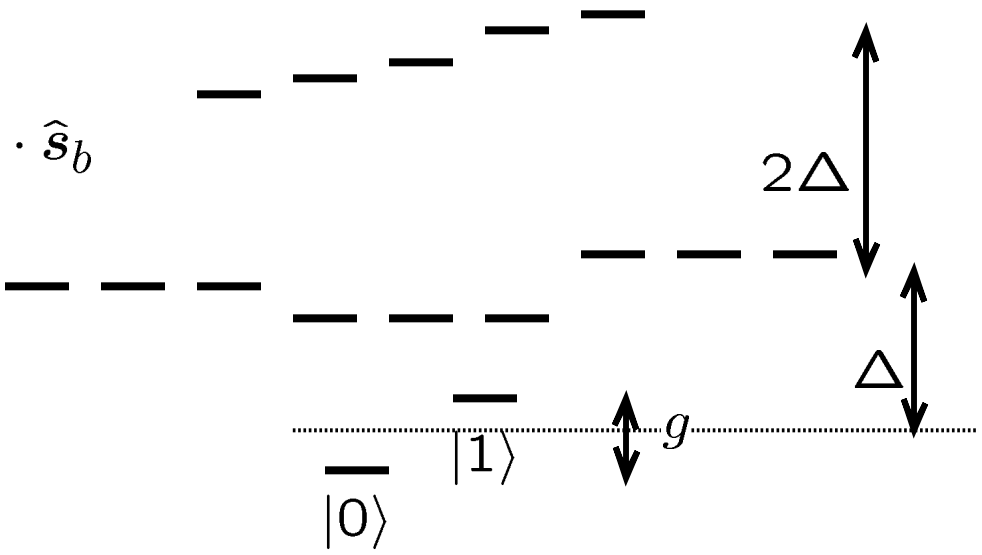
Bacon, Brown, and Whaley, **PRL** 87, 247902 (200

Computing on Supercoherence

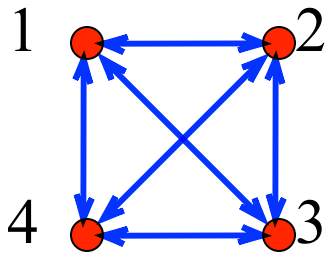
Manipulation of the encoded qubit can be enacted by additional exchange energy:

$$\mathbf{H} = \frac{\Delta}{2} \sum_{i,j=1}^{4} \hat{\boldsymbol{s}}_i \cdot \hat{\boldsymbol{s}}_j + g \hat{\boldsymbol{s}}_a \cdot \hat{\boldsymbol{s}}_b$$

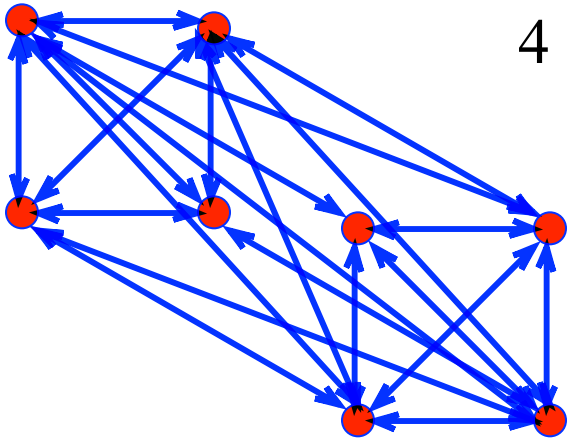
$g \ll T \Rightarrow$ supercoherence property not lost.



Extra exchange on



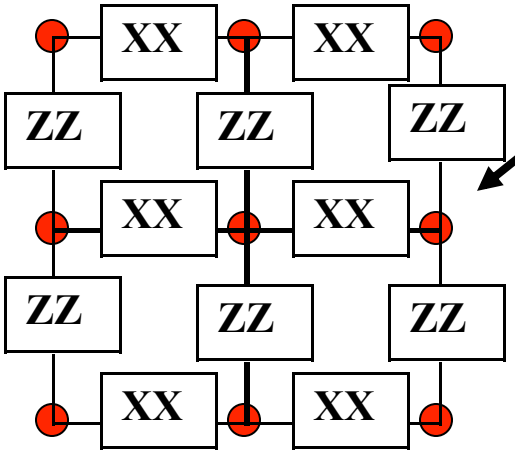
gives all SU(2) on encoded qubit



To compute on between two encoded qubits exchange must be turned on between eight physical qubits

Beyond Supercoherence

Supercoherence is small step towards the ultimate goal of natural fault tolerant quantum computation (it is physical error detection.)

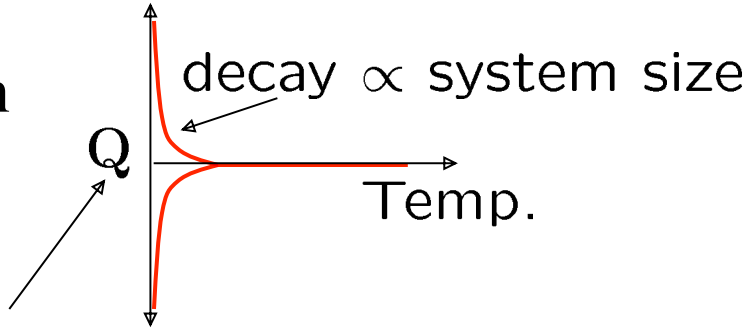


This systems ground state is a quantum error correcting code.

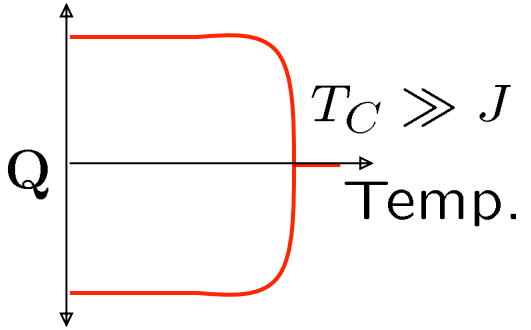
Kitaev and Freedman's topological codes are examples of systems whose energy eigenstates are quantum error correcting code states.

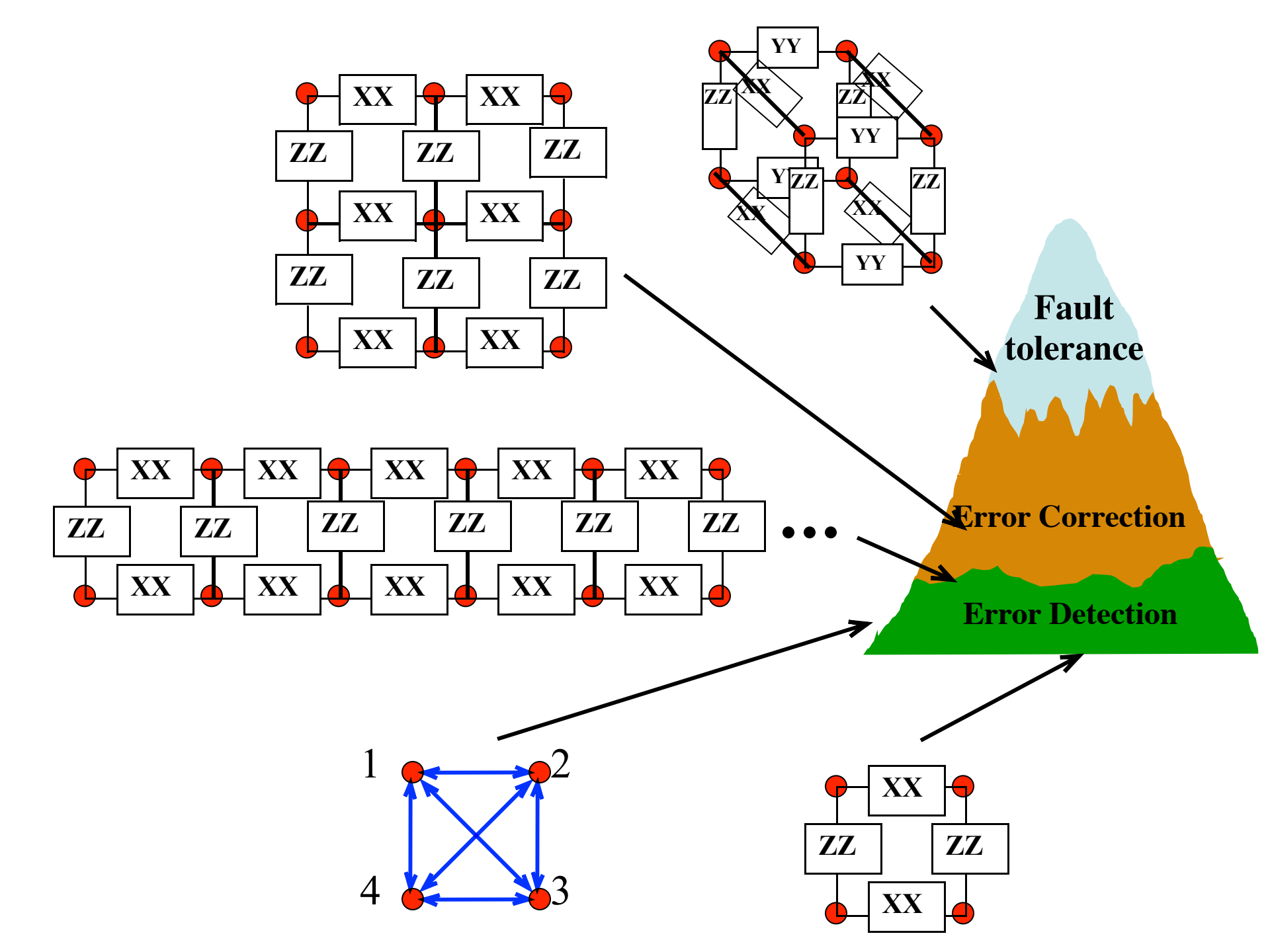
Problem: find a system with a persistent coherence order parameter

All known systems



The goal




XX
ZZ
YY
Fault tolerance
Error Correction
Error Detection
1
2
3
4

Quantum Computer Revolution = (Interesting!) Physics?

DO THERE EXIST SYSTEMS WHOSE PHYSICS GUARANTEES ROBUST QUANTUM COMPUTATION?

Collaborators: K. Birgitta Whaley and K. R. Brown
Department of Chemistry
U.C. Berkeley